

7) Inhomogeneous Solution: Wave Propagation in Media

In the previous section, we examined electromagnetic radiation, governed by an inhomogeneous wave equation arising from time-varying charge and current distributions. Building upon this foundation, we now turn to the study of wave propagation in different materials.

When an electromagnetic wave interacts with matter, the induced polarization and magnetization act as secondary sources, leading to another class of inhomogeneous wave equations. Solving these equations not only elucidates the origin of the refractive index but also unveils the fundamental mechanism behind optical dispersion. Through this analysis, we establish a profound connection between microscopic material response—dictated by atomic and molecular dynamics—and macroscopic electromagnetic phenomena, demonstrating how intrinsic material properties shape wave propagation.

At first glance, the refractive index might seem like a simple and familiar concept—after all, its basic definition is introduced in introductory physics. However, a deeper examination reveals profound and subtle physics underlying this seemingly straightforward quantity. Consider the following fundamental questions:

(i) Macroscopic vs. Microscopic Interpretation:

In water or glass, the refractive index $n > 1$ implies that the phase velocity of light is reduced. Yet microscopically, light propagation in a medium consists of nothing more than the emission and reabsorption of radiation by induced dipoles (molecules or atoms). Why should the collective response of these dipoles result in an effective slowdown of light?

(ii) Dispersion and Newton's Prism:

As demonstrated by Newton's prism experiment, blue light refracts more strongly than red light. What microscopic mechanisms give rise to this wavelength-dependent behavior, and how does it connect to the frequency response of the medium?

(iii) Opacity and Transparency Across the Spectrum:

Why are polished metals highly reflective in the visible range, while most materials—including metals—become transparent to X-rays? What determines the transition between these regimes?

(iv) Beyond Conventional Refractive Indices:

The refractive index is typically greater than one, but can it be less than one? Can it be zero or even negative? Under what conditions does it become complex, and what physical phenomena does this signify?

These questions illustrate that our understanding of the refractive index is far from trivial. Its full

description requires a synthesis of classical electrodynamics, quantum mechanics, and material physics, revealing rich and sometimes counterintuitive behavior.

In developing the theory of the refractive index, we focus primarily on interactions mediated by electric dipole moments, as these dominate the electromagnetic response of most dielectric materials at optical frequencies. The induced polarization of bound charges—manifesting as oscillating electric dipoles—accounts for the majority of the refractive index’s magnitude in conventional media.

A distinct regime arises in magnetic dipole interactions, such as those exploited in magnetic resonance imaging (MRI). Here, electromagnetic waves in the radiofrequency/microwave band couple to nuclear or electronic magnetic moments (e.g., spin-flip transitions in nuclei). However, such magnetic responses are negligible at optical frequencies, where the electric dipole term overwhelmingly governs the material’s behavior.

For most materials across the electromagnetic spectrum—particularly in the optical regime—the relative permeability μ_r is exceedingly close to unity ($\mu_r \approx 1$). This arises because the magnetic susceptibility of non-magnetic dielectrics is vanishingly small at high frequencies; atomic or molecular magnetic moments cannot respond appreciably to the rapidly oscillating fields of visible or higher-frequency radiation. Consequently, in optical and ultraviolet regimes, it is both practical and physically justified to assume $\mu_r = 1$, simplifying the description of wave propagation to a focus on electric polarization.

While we focus exclusively on the electric response of the material, it is important to recognize that the simple static relation

$$\vec{P} = \epsilon_0 \chi_e \vec{E}$$

fails to describe the true dynamical behavior of the system. This expression holds only in the static limit, where the polarization instantaneously follows the applied field.

In the dynamic regime—where the material’s response remains linear (assuming the field amplitude is not too large)—the polarization $\vec{P}(t)$ must instead be treated as the causal output of the material under the time-dependent excitation $\vec{E}(t)$. The correct description requires a convolution integral:

$$\vec{P}(t) = \epsilon_0 \int_{-\infty}^t \chi_e(t - \tau) \vec{E}(\tau) d\tau,$$

where the electric susceptibility $\chi_e(t - \tau)$ acts as the response function of the medium.

① The Lorentz model of dielectric materials

For simplicity, consider the interaction of an electromagnetic wave with a linear, isotropic, and homogeneous material. The electric field oscillates in the x-direction as $e^{-j\omega t}$ and propagates along the +z-direction. Due to the isotropy of the material, the induced polarization $\vec{P}$ also oscillates in the x-direction. Moreover, because the material is linear, $\vec{P}$ varies with the same temporal dependence $e^{-j\omega}$, and no new frequencies are generated. We will begin by considering a dielectric material and subsequently extend our theory to include metals. In all cases, the material is assumed to be non-magnetic, i.e., with relative permeability $\mu_r = 1$.

We begin our derivation with the macroscopic form of Maxwell's equations. Taking the curl of both sides of the Maxwell-Faraday equation,

$$\nabla \times (\nabla \times \vec{E}) = -\frac{\partial}{\partial t} (\nabla \times \vec{B}),$$

and using $\vec{B} = \mu_0 \vec{H}$, we obtain

$$\nabla \times (\nabla \times \vec{E}) = -\mu_0 \frac{\partial}{\partial t} (\nabla \times \vec{H}).$$

Assuming there is no free current within the material, we apply Ampere's Law with Maxwell's addition,

$$\nabla \times \vec{H} = \partial_t \vec{D},$$

which leads to

$$\nabla \times (\nabla \times \vec{E}) = -\mu_0 \partial_{tt} \vec{D} = -\mu_0 \partial_{tt} (\epsilon_0 \vec{E} + \vec{P}).$$

Using the identity $\nabla \times (\nabla \times \vec{E}) = \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E}$, we arrive at

$$\nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\mu_0 \epsilon_0 \partial_{tt} \vec{E} - \mu_0 \partial_{tt} \vec{P}.$$

When deriving the homogeneous wave equation in vacuum, we set $\nabla \cdot \vec{E} = 0$ due to the absence of any charge. However, in the present context, we must proceed with caution, as the charge density is not zero. Instead, it arises from bound charges:

$$\nabla \cdot \vec{E} = \frac{\rho_b}{\epsilon_0} = -\frac{\nabla \cdot \vec{P}}{\epsilon_0}.$$

Recall that we have assumed the polarization $\vec{P}$ takes the form of a plane wave propagating along the +z-direction and oscillating along the x-direction. Therefore, $P_y = P_z = 0$, and by the plane wave property, $\vec{P}$ does not vary with the x-coordinate, implying $\partial_x \equiv 0$. As a result,

$$\partial_x P_x + \partial_y P_y + \partial_z P_z = 0,$$

which means the divergence of $\vec{P}$ vanishes, and thus

$$\nabla \cdot \vec{E} = -\frac{\nabla \cdot \vec{P}}{\epsilon_0} = 0.$$

In conclusion, we obtain the inhomogeneous wave equation that governs the propagation of electromagnetic waves through a dielectric material:

$$\boxed{\nabla^2 \vec{E} - \frac{1}{c^2} \partial_{tt} \vec{E} = \mu_0 \partial_{tt} \vec{P}}.$$

It is important to note that we cannot directly apply

$$\vec{P} = \varepsilon_0 \chi_e \vec{E}$$

as this relation assumes that the polarization responds instantaneously to the applied electric field—an assumption that is physically unrealistic. Instead, we proceed by substituting a set of trial plane-wave solutions into the wave equation:

$$\vec{E} = \tilde{E}_x e^{j(kz - \omega t)} \hat{x},$$

$$\vec{P} = \tilde{P}_x e^{j(kz - \omega t)} \hat{x}.$$

Inserting these expressions into the wave equation, we obtain:

$$\partial_{zz} \tilde{E}_x e^{j(kz - \omega t)} - \frac{1}{c^2} \partial_{tt} \tilde{E}_x e^{j(kz - \omega t)} = \mu_0 \partial_{tt} \tilde{P}_x e^{j(kz - \omega t)}.$$

This simplifies to the algebraic equation:

$$\boxed{-k^2 \tilde{E}_x(\omega) + \frac{\omega^2}{c^2} \tilde{E}_x(\omega) = -\mu_0 \omega^2 \tilde{P}_x(\omega)}.$$

Here, we explicitly express the complex amplitudes $\tilde{E}_x$ and $\tilde{P}_x$ as functions of the angular frequency ω , to emphasize their frequency dependence. This formulation also reflects the fact that a convolution in the time domain becomes a simple multiplication in the frequency domain:

$$\tilde{P}_x(\omega) = \varepsilon_0 \tilde{\chi}_e(\omega) \tilde{E}_x(\omega).$$

Substituting this relation into the frequency-domain wave equation yields

$$-k^2 \tilde{E}_x(\omega) + \frac{\omega^2}{c^2} \tilde{E}_x(\omega) = -\varepsilon_0 \mu_0 \omega^2 \tilde{\chi}_e(\omega) \tilde{E}_x(\omega).$$

Cancelling the common factor $\tilde{E}_x(\omega)$ from both sides and introducing the vacuum wavenumber $k_0 = \omega/c$, we obtain

$$k^2 = \frac{\omega^2}{c^2} (1 + \tilde{\chi}_e(\omega)) = k_0^2 \tilde{n}^2,$$

where

$$\boxed{\tilde{n}^2(\omega) = 1 + \tilde{\chi}_e(\omega) = \tilde{\varepsilon}_r(\omega)},$$

and $\tilde{n}(\omega)$ is the frequency-dependent complex refractive index.

At the most fundamental level, material polarization arises from induced molecular dipoles. When an external electric field oscillates, these dipole moments are driven to oscillate at the same

frequency, each acting as a microscopic source of dipole radiation. The total electromagnetic wave propagating through the material is the result of the superposition of the secondary waves emitted by the ensemble of dipoles and the original incident wave. The net effect is that the electromagnetic wave appears to slow down as it travels through the medium. This remarkable phenomenon is rigorously explained by the Ewald–Oseen extinction theorem, which lies beyond the scope of this course (for further details, refer to classical texts such as Born and Wolf’s *Principles of Optics*). Our specific goal here is to determine the frequency response of the dielectric material, expressed by the electric susceptibility $\tilde{\chi}_e(\omega)$. The figure below illustrates how a molecular dipole responds to an external oscillating electric field.

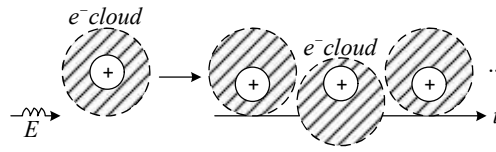


Figure. An oscillating dipole induced by an external electric field.

The attraction force between the positive ion and the electron cloud can be modeled as a spring mass system, whose oscillation is driven by the external electric field. The figure below illustrates this model.

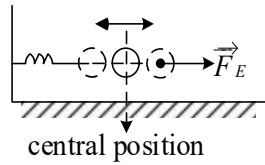


Figure. The spring–mass system representing the Lorentz oscillator model.

In the figure, the spring represents the molecular restoring force that binds the electron to its equilibrium position, while $\vec{F}_E$ denotes the external driving force exerted by the electric field of the incident light. We now examine the motion of the electron under a single driving frequency.

According to Newtonian mechanics, the motion of the electron follows the differential equation:

$$m\ddot{\tilde{r}}(t) = -m\gamma\dot{\tilde{r}}(t) - m\omega_0^2\tilde{r}(t) - q_e\tilde{E}(t),$$

where m and $-q_e$ are the mass and charge of the electron, respectively, and $\tilde{r}$ denotes the displacement from the equilibrium position. The displacement is expressed as a complex quantity to reflect the harmonic nature of the motion: $\tilde{r}(t) = \tilde{r}(\omega)e^{-j\omega t}$, analogous to the complex representation of the oscillating electric field $\tilde{E}(t) = \tilde{E}(\omega)e^{-j\omega t}$. The damping coefficient is given

by $\gamma = 1/\tau$, where τ is the average time between collisions, and $m\omega_0^2$ corresponds to the classical spring constant, with ω_0 being the natural (resonance) frequency of the oscillator.

The left-hand side of the equation represents the inertial term (mass times acceleration), while the three terms on the right-hand side correspond to the damping (frictional) force, the restoring (elastic) force, and the external driving force from the electric field, respectively. This equation is a direct application of Newton's second law of motion.

The above equation is converted into the following algebraic equation when we substitute $\tilde{r}(t) = \tilde{r}(\omega)e^{-j\omega t}$ and $\tilde{E}(t) = \tilde{E}(\omega)e^{-j\omega t}$:

$$-\omega^2 m \tilde{r}(\omega) - j\gamma m \omega \tilde{r}(\omega) + m\omega_0^2 \tilde{r}(\omega) = -q_e \tilde{E}(\omega).$$

Solving for $\tilde{r}(\omega)$, we find:

$$\tilde{r}(\omega) = -\frac{q_e}{-m\omega^2 - j\gamma m \omega + m\omega_0^2} \tilde{E}(\omega).$$

The dipole moment of a single oscillating electron is given by:

$$\tilde{\mu}_{\text{dipole}}(\omega) = -q_e \tilde{r}(\omega) = \underbrace{\left(\frac{q_e^2/m}{-\omega^2 - j\gamma \omega + \omega_0^2} \right)}_{\tilde{\beta}(\omega): \text{polarizability}} \tilde{E}(\omega),$$

Multiplying by the number density N of such dipoles, the macroscopic polarization becomes:

$$\tilde{P}(\omega) = N \tilde{\mu}(\omega) = N \tilde{\beta}(\omega) \tilde{E}(\omega).$$

The electric susceptibility is defined as:

$$\tilde{\chi}_e(\omega) = \frac{\tilde{P}(\omega)}{\epsilon_0 \tilde{E}(\omega)} = \frac{N}{\epsilon_0} \tilde{\beta}(\omega),$$

which leads to:

$$\tilde{\chi}_e(\omega) = \frac{N q_e^2}{\epsilon_0 m} \cdot \frac{1}{-\omega^2 - j\gamma \omega + \omega_0^2}.$$

Define the plasma frequency ω_p , which is an important material-dependent parameter, as:

$$\omega_p = \sqrt{\frac{N q_e^2}{\epsilon_0 m}},$$

and recalling that $\tilde{n}^2(\omega) = 1 + \tilde{\chi}_e(\omega)$, we finally conclude that

$$\tilde{n}^2(\omega) = 1 + \frac{\omega_p^2}{-\omega^2 - j\gamma \omega + \omega_0^2}.$$

Notes:

- (i) The model described above is known as the Lorentz model for dielectric response. Despite its simplicity, it provides remarkably accurate predictions across many materials.
- (ii) The frequency-dependent relative permittivity is given by $\tilde{\epsilon}_r(\omega) = \tilde{n}^2(\omega)$, which is both

complex and dispersive. Therefore, the complex refractive index can be written as:

$$\tilde{n}(\omega) = n(\omega) + j\kappa(\omega),$$

where $\kappa(\omega)$ is the extinction coefficient.

The electric field of a plane wave propagating in the z-direction in such a medium is:

$$\tilde{E} = \tilde{E}_0 e^{j(k_0 \tilde{n} z - \omega t)} = \tilde{E}_0 e^{-k_0 \kappa z} e^{j(k_0 n z - \omega t)}.$$

The exponential decay factor $e^{-k_0 \kappa z}$ reflects the attenuation of the wave due to absorption.

The intensity of the wave is proportional to $|\tilde{E}|^2$, and therefore decays as:

$$\tilde{E} \tilde{E}^* \propto e^{-2k_0 \kappa z} = e^{-\alpha z},$$

where the attenuation coefficient is defined as:

$$\boxed{\alpha = 2\kappa k_0}.$$

(iii) The frequency-dependent behavior of several key optical properties is illustrated in the figure below, assuming the parameters $\omega_p = 4, \omega_0 = 2, \gamma = 0.1$. These values are chosen for illustrative purposes only and do not correspond to any specific material. The plots include the real and imaginary parts of the refractive index, the real and imaginary parts of the relative permittivity, as well as its magnitude and phase angle, and finally the reflectance at normal incidence. While the absolute values are not physically realistic, the qualitative trends and features of these curves provide meaningful insights into the dispersive and absorptive behavior of dielectric materials.

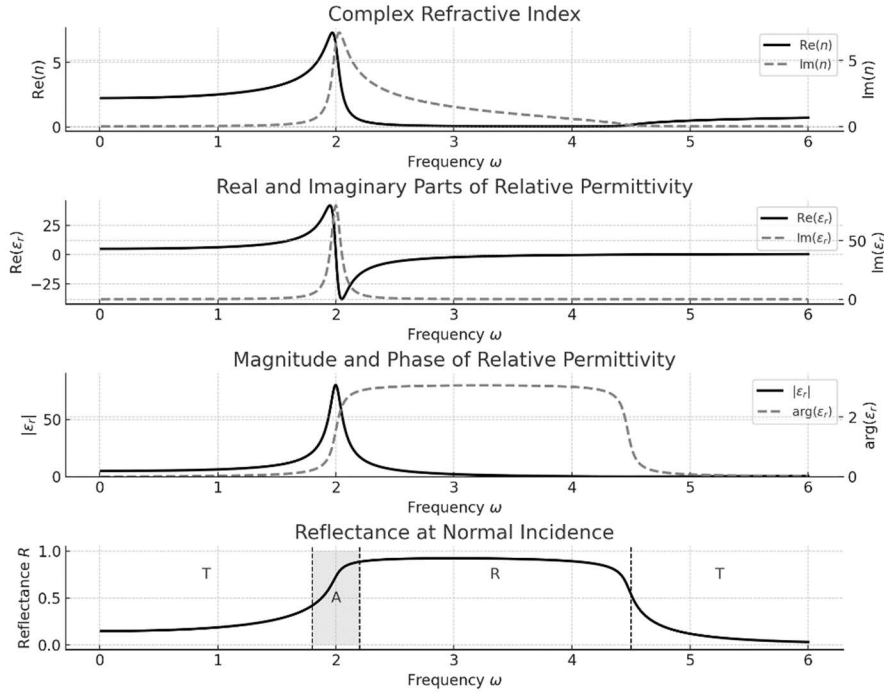


Figure. Illustrative plots of the complex refractive index, relative permittivity and reflectance as functions of frequency.

The reflectance shown in the last subplot is calculated using the Fresnel formula at normal incidence,

$$R = \frac{(n - 1)^2 + \kappa^2}{(n + 1)^2 + \kappa^2},$$

This expression, whose derivation will be provided in the next lecture, is especially useful because it relates a directly measurable quantity—reflectance—to the material's intrinsic optical constants. As such, it serves as a practical tool for evaluating dispersion and absorption characteristics, and for probing electronic resonance features through optical measurements.

In the last subplot, the labels T, A, R, T in the reflectance plot denote four distinct regimes of wave-material interaction as the frequency of the incident electromagnetic wave varies. At low frequencies (T, transmission), the material's polarization can follow the field adiabatically, resulting in a real refractive index and low reflectance, so the wave propagates through the medium with minimal loss. As the frequency approaches the material's resonance frequency ω_0 , the system enters the A (absorption) zone. Here, the driving frequency matches the natural oscillation of bound electrons, leading to strong energy absorption, a peak in the imaginary part of the permittivity, and attenuation of the wave. Beyond the resonance but still below the plasma frequency ω_p , in the R (reflection) regime, the real part of the permittivity becomes negative, causing the refractive index to become purely imaginary or complex with a large imaginary component. This leads to strong reflection and suppression of wave propagation. At even higher frequencies ($\omega > \omega_p$), the inertia of bound charges prevents them from responding to the rapidly oscillating field, restoring a positive real permittivity and allowing waves to transmit again—this is the second T (transmission) zone.

Normal and anomalous dispersion refer to how the real part of the refractive index changes with frequency: in normal dispersion, the refractive index increases with frequency, whereas in anomalous dispersion, it decreases with frequency. In the first subplot of the figure, this behavior can be observed as $\text{Re}(n)$ increases at low and high frequencies (normal dispersion), but decreases in a narrow frequency range near the resonance frequency ω_0 , indicating anomalous dispersion induced by the material's resonant response.

② The Drude model for conductors

In the Lorentz model, the square of the complex refractive index is expressed as

$$\tilde{n}^2(\omega) = 1 + \frac{\omega_p^2}{-\omega^2 - j\gamma\omega + \omega_0^2},$$

where the term ω_0^2 on the denominator is associated with the restoring force that tries to return displaced charges to equilibrium. In metals, however, a significant portion of the electrons are free,

meaning they are not bound to atoms and experience no restoring force. For these free electrons, the restoring term vanishes, and we set $\omega_0 = 0$. This simplifies the expression to:

$$\tilde{n}^2(\omega) = 1 + \frac{\omega_p^2}{-\omega^2 - j\gamma\omega} = 1 - \frac{\omega_p^2}{\omega^2 + j\gamma\omega}$$

This modified model is known as the Drude model, which effectively describes the optical response of metals dominated by free electron dynamics.

***Extra explanation of the model:**

Before studying the Drude model in detail, let us pause to revisit the validity of the formulation.

Recall that the complex refractive index was defined via the relation

$$\tilde{n}^2(\omega) = 1 + \tilde{\chi}_e(\omega) = \tilde{\epsilon}_r(\omega),$$

This relation emerged from solving the inhomogeneous wave equation in a dielectric medium, where we used the key assumption:

$$\nabla \cdot \vec{E} = -\frac{\nabla \cdot \vec{P}}{\epsilon_0} = 0.$$

Such an assumption is valid only in dielectric materials, where polarization arises from bound charges. However, we have not yet shown that metals, which support free currents instead of bound charges, admit the same form of the wave equation, or that the definition of complex permittivity remains meaningful in this context.

To justify this, we re-express Maxwell's equations in a conductor. Starting from the curl of Faraday's law:

$$\nabla \times \nabla \times \vec{E} = -\mu_0 \frac{\partial}{\partial t} \nabla \times \vec{H} = -\mu_0 \frac{\partial}{\partial t} \left(\vec{J}_f + \epsilon_0 \frac{\partial}{\partial t} \vec{E} \right),$$

and applying the vector identity $\nabla \times \nabla \times \vec{E} = \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E}$, we obtain

$$\nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\mu_0 \frac{\partial}{\partial t} \vec{J}_f - \epsilon_0 \mu_0 \frac{\partial^2}{\partial t^2} \vec{E}.$$

Now, move to the frequency domain by assuming time-harmonic fields, assuming plane-wave

fields such as $\vec{E} = \tilde{E}(\omega) e^{j(kz - \omega t)} \hat{x}$, and write the equation as

$$\nabla \left(\frac{\tilde{\rho}}{\epsilon_0} \right) + k^2 \tilde{E} = j\omega \mu_0 \tilde{J}_f + \frac{\omega^2}{c^2} \tilde{E}.$$

To simplify, we apply the continuity equation in the frequency domain:

$$\nabla \cdot \tilde{J}_f = j\omega \tilde{\rho}.$$

This implies that if

$$\nabla \cdot \tilde{J}_f = 0,$$

then $\tilde{\rho} = 0$, and hence $\nabla \cdot \vec{E} = 0$. This is a reasonable assumption for plane-wave excitation in homogeneous media, and is the same logic we used when proving $\nabla \cdot \vec{P} = 0$ in dielectrics.

Assuming Ohm's law in the frequency domain:

$$\tilde{\vec{J}}_f = \tilde{\sigma}(\omega)\tilde{\vec{E}},$$

and considering a plane wave polarized in the x-direction and propagating along +z, we obtain:

$$-k^2\tilde{E}_x(\omega) + \frac{\omega^2}{c^2}\tilde{E}_x(\omega) = -j\omega\mu_0\tilde{\sigma}(\omega)\tilde{E}_x(\omega).$$

This equation has the same structure as the one previously derived for dielectrics:

$$-k^2\tilde{E}_x(\omega) + \frac{\omega^2}{c^2}\tilde{E}_x(\omega) = -\varepsilon_0\mu_0\omega^2\tilde{\chi}_e(\omega)\tilde{E}_x(\omega).$$

Both can be rearranged into the common form:

$$(-k^2 + k_0^2\tilde{n}^2(\omega))\tilde{E}_x(\omega) = 0,$$

with the distinction that: for dielectrics, $\tilde{n}^2(\omega) = 1 + \tilde{\chi}_e(\omega)$, for conductors, $\tilde{n}^2(\omega) = 1 + \frac{j\tilde{\sigma}(\omega)}{\varepsilon_0\omega}$.

Substituting the Drude-model conductivity $\tilde{\sigma}(\omega) = \frac{\varepsilon_0\omega_p^2}{\gamma - j\omega}$ into the above gives:

$$\tilde{n}^2(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + j\gamma\omega},$$

which is precisely the Drude expression for conductors.

Notes: (i) **An illustrative plot** with $\omega_p = 3$, $\omega_0 = 0$, $\gamma = 0.1$:

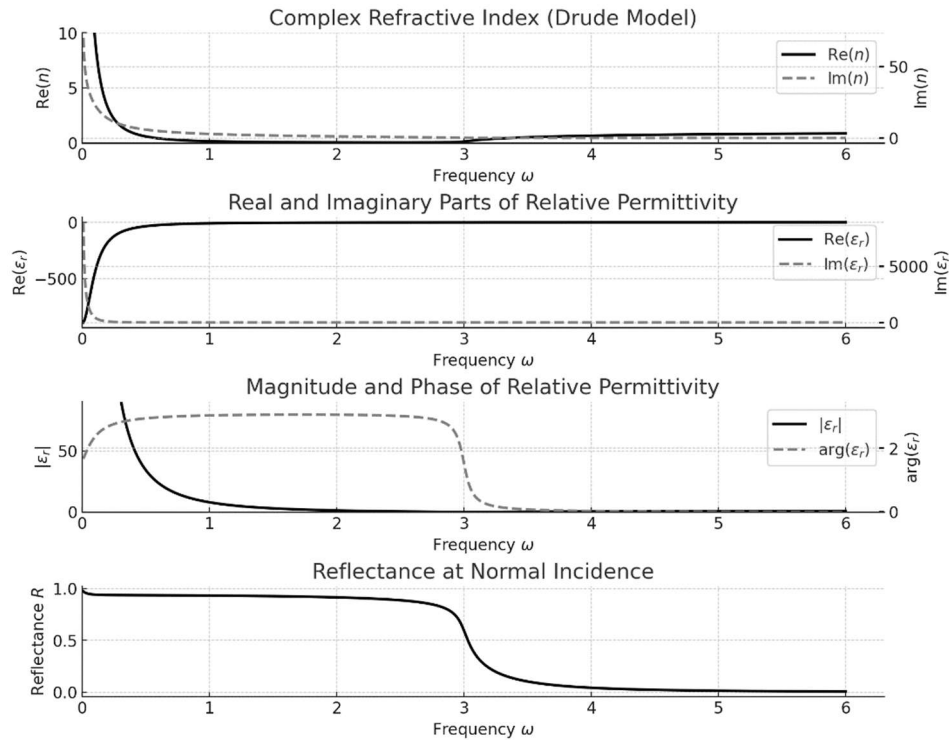


Figure. Illustrative plots for a conductor.

(ii) **The plasma frequency** $\omega_p = \sqrt{\frac{Nq_e^2}{\epsilon_0 m}}$ is an important parameter for metals, representing the natural oscillation frequency of the free electron gas in response to an external electric field. Physically, it corresponds to the collective oscillation of conduction electrons against the fixed positive ion background. When the incident light frequency ω is below ω_p , the real part of the metal's permittivity becomes negative, leading to strong reflection and the material behaves as an opaque mirror. Conversely, for $\omega > \omega_p$, the permittivity becomes positive and the metal can support electromagnetic wave propagation, becoming partially transparent. The plasma frequency thus defines a sharp transition between reflective and transmissive regimes and plays a key role in determining a metal's optical and electronic behavior.

***Extra explanation of why the Plasma Frequency is the Natural Oscillation Frequency of Free Electrons**

Consider a metal composed of free electrons with number density N , each with charge $-q_e$ and mass m . Let the electrons be displaced by a small amount $s(x,t)$ along the x-direction, while the positive ion background remains fixed. If the displacement s is spatially non-uniform, a local charge imbalance appears due to divergence in the electron density. The resulting net charge density is approximately:

$$\rho = Nq_e \frac{\partial s}{\partial x}.$$

From Gauss's law in 1D,

$$\partial_x E_x = \frac{\rho(x)}{\epsilon_0} = \frac{Nq_e}{\epsilon_0} \frac{\partial s}{\partial x}.$$

Integrating with respect to x , and assuming $E_x = 0$ when $s = 0$, we get:

$$E_x(x) = \frac{Nq_e}{\epsilon_0} s(x).$$

This electric field acts as a restoring force on the displaced electrons. The equation of motion for an electron under this field is:

$$m \frac{d^2 s}{dt^2} = -q_e E_x = -\frac{Nq_e^2}{\epsilon_0} s.$$

Rewriting:

$$\frac{d^2 s}{dt^2} + \omega_p^2 s = 0,$$

where the plasma frequency is defined as:

$$\omega_p = \sqrt{\frac{Nq_e^2}{\epsilon_0 m}}.$$

The solution to the above equation has the form $s(t) \sim e^{-j\omega_p t}$. Thus, ω_p is the natural frequency of longitudinal oscillation of the electron gas relative to the fixed positive background. These oscillations are known as plasma oscillations, and the electrons collectively respond as a compressible fluid under electrostatic restoring forces.

Example:

For copper, assume one electron per atom, the relevant parameters are:

$$\begin{cases} N = 8.5 \times 10^{28} \text{ m}^{-3}, & q_e = 1.6 \times 10^{-19} \text{ C}, & \epsilon_0 = 8.85 \times 10^{-12} \text{ F} \cdot \text{m}^{-1} \\ m = 0.11 \times 10^{-31} \text{ kg} & \text{(electron's effective mass).} \\ \tau = 2.24 \times 10^{-14} \text{ s} & \text{(mean time between collisions)} \\ 1/\tau = 4 \times 10^{13} \text{ s}^{-1} \end{cases}$$

Using these values, the plasma frequency of copper is calculated to be

$$\omega_p = 16.4 \times 10^{15} \text{ s}^{-1}.$$

This is not a bad estimation, which is reasonably close to the real value of $13.4 \times 10^{15} \text{ s}^{-1}$. Such a plasma frequency corresponds to a wavelength of 115 nm, which is in the ultra-violet range.

(iii) High-frequency limit (ω comparable to ω_p)

In the Drude model, the complex refractive index $\tilde{n}(\omega)$ of a free-electron metal is related to the angular frequency by

$$\tilde{n}^2(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + j\gamma\omega}.$$

In the high-frequency limit where $\omega \gg \gamma$, the expression simplifies to:

$$\tilde{n}^2(\omega) \approx 1 - \frac{\omega_p^2}{\omega^2}.$$

This leads to two distinct regimes:

a. For $\omega > \omega_p$:

$$\tilde{n}^2(\omega) > 0, 0 < \tilde{n} < 1.$$

In this regime, $\tilde{n}$ is purely real, and the phase velocity exceeds the speed of light in vacuum ($v_p = \frac{c}{\tilde{n}} > c$). The material behaves like a highly transmissive dielectric, allowing electromagnetic waves to propagate with minimal attenuation.

b. For $\omega < \omega_p$:

$$\tilde{n}^2(\omega) < 0, \text{ so } \tilde{n} = j\kappa,$$

where κ is real and positive. Here, $\tilde{n}$ is purely imaginary, indicating an evanescent wave that decays exponentially within the material. Although this behavior suggests the medium is absorptive, a more careful analysis reveals otherwise.

To evaluate the reflectivity, consider the Fresnel reflection coefficient at normal incidence:

$$R = \frac{(n - 1)^2 + \kappa^2}{(n + 1)^2 + \kappa^2}.$$

In the frequency range $\gamma \ll \omega < \omega_p$, $\tilde{n}$ is almost purely imaginary and its real part $n \approx 0$, the above expression simplifies to

$$R \approx \frac{1 + \kappa^2}{1 + \kappa^2} = 1.$$

Thus, despite the appearance of absorption, the material is nearly perfectly reflective in this regime. This explains why metals exhibit strong reflection for incident light with frequencies below the plasma frequency.

Example 1: Copper's ω_p is on the order of 10^{16} sec^{-1} , corresponding to $\lambda_p \sim 100 \text{ nm}$.

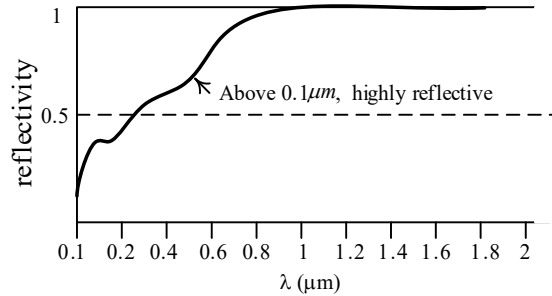


Figure. Reflectivity of copper as a function of wavelength.

From the figure above, we see that the experimentally-measured reflectivity of copper is in very good agreement with the theoretical prediction.

Example 2: Earth's ionosphere:

The Earth's ionosphere, located approximately 60 km to 1000 km above the surface, contains a significant concentration of free electrons. A typical electron density in this region is $N \approx 10^{12} \text{ m}^{-3}$, this corresponds to a plasma frequency of $\omega_p = 56 \times 10^6 \text{ s}^{-1} \approx 2\pi \times 9 \text{ MHz}$.

Electromagnetic waves interact with the ionospheric plasma depending on their frequency relative to the plasma frequency: For frequencies above 9 MHz, the wave can propagate through the ionosphere and escape into space. For frequencies below 9 MHz, the wave is reflected back toward the Earth's surface.

This phenomenon explains why AM radio signals, which typically operate at frequencies between 0.5 MHz to 1.7 MHz, can be received over large distances—well beyond the line of sight—via reflection from the ionosphere.

(iv) Low frequency limit ($\omega \ll \omega_p, \omega \ll \gamma$)

We begin with the Drude model expression for the complex refractive index:

$$\tilde{n}^2(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + j\gamma\omega}.$$

In the low-frequency limit, both $\omega \ll \omega_p$ and $\omega \ll \gamma = 1/\tau$, so we may neglect ω^2 in the denominator:

$$\tilde{n}^2(\omega) \approx 1 - \frac{\omega_p^2}{j\gamma\omega} = 1 + j \frac{\omega_p^2}{\gamma\omega}.$$

Since $\frac{\omega_p^2}{\gamma\omega} \gg 1$, we can further approximate:

$$\tilde{n}^2(\omega) \approx j \frac{\omega_p^2}{\gamma\omega}.$$

Invoking our definition of the complex refractive index $\tilde{n} = n + j\kappa$, and solve for $\tilde{n}$ by taking the square root:

$$\tilde{n}(\omega) \approx \sqrt{j \frac{\omega_p^2}{\gamma\omega}} = \frac{\omega_p}{\sqrt{2\gamma\omega}} (1 + j).$$

Hence, the real and imaginary parts are equal: $n = \kappa = \frac{\omega_p}{\sqrt{2\gamma\omega}}$, and the wave is strongly attenuated in the medium.

The attenuation coefficient α (in units of m^{-1}) is related to the imaginary part κ by:

$$\alpha = 2k_0\kappa = \frac{2\omega}{c} \frac{\omega_p}{\sqrt{2\gamma\omega}} = \frac{\omega_p}{c} \sqrt{\frac{2\omega}{\gamma}} = \frac{\omega_p}{c} \sqrt{2\omega\tau}.$$

Let us now evaluate this for copper, with typical parameters:

Plasma frequency: $\omega_p \approx 1.6 \times 10^{16} \text{ s}^{-1}$,

Relaxation time: $\tau \approx 2.4 \times 10^{-14} \text{ s}$,

Frequency: $f = 100 \text{ MHz}$, i.e., $\omega = 2\pi \times 10^8 \text{ rad/s}$.

Substituting these values:

$$\alpha \approx \frac{1.6 \times 10^{16}}{3 \times 10^8} \cdot \sqrt{2 \times 2\pi \times 10^8 \times 2.4 \times 10^{-14}} \approx 3 \times 10^5 \text{ m}^{-1}.$$

The corresponding penetration depth is

$$\delta = \frac{1}{\alpha} \approx \frac{1}{3 \times 10^5} \text{ m} \approx 3 \text{ } \mu\text{m}.$$

Compare this to the wavelength at 100 MHz,

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10^8} = 3 \text{ m}.$$

Thus, the wave only penetrates about one-millionth of a wavelength into the conductor:

$$\delta \approx \frac{\lambda}{10^6}.$$

This behavior is consistent with the DC limit, in which static or slowly varying fields are expelled from the interior of a good conductor due to the formation of surface currents. Despite the large imaginary part of the refractive index (which might naively suggest absorption), the reflectivity remains close to unity, as discussed earlier. Therefore, **most of the incident energy is reflected**, not absorbed.

8) Phase and Group Velocity in Dispersive Media

Consider a monochromatic plane wave of the form

$$\tilde{E}(z, t) = \tilde{E}_0 e^{i(kz - \omega t)}.$$

The phase velocity v_p is defined as the speed at which a point of constant phase (e.g., a wave crest) propagates:

At constant phase, $kz - \omega t = \text{const}$, if we take the time derivative of both sides:

$$k \frac{dz}{dt} - \omega = 0,$$

$\frac{dz}{dt}$ is the speed this constant phase (e.g., the wave crest) propagates along the +z direction, as a result,

$$v_p = \frac{dz}{dt} = \frac{\omega}{k}.$$

In a medium with refractive index $n(\omega)$, the wave number is related to the free-space wave number $k_0 = \omega/c$ by:

$$k = n(\omega)k_0 = \frac{n(\omega)\omega}{c},$$

Thus, the phase velocity becomes:

$$v_p = \frac{\omega}{k} = \frac{c}{n(\omega)}.$$

If $n > 1$, then $v_p < c$. However, in some frequency ranges—notably for metals at frequencies just above the plasma frequency ω_p —the refractive index satisfies $0 < n(\omega) < 1$, resulting in $v_p > c$. This superluminal phase velocity does not imply that information or energy is traveling faster than the speed of light.

Real signals cannot be purely monochromatic; instead, they are modulated and have a finite bandwidth. Such signals must be represented as a superposition of frequencies. The electric field of a modulated wave packet is expressed as:

$$E(t, z) = \int_{-\infty}^{\infty} \tilde{E}(\omega) e^{j[k(\omega)z - \omega t]} d\omega,$$

where $k(\omega)$ is frequency-dependent due to dispersion.

To analyze signal propagation, expand $k(\omega)$ around the central frequency ω_0 using a Taylor series:

$$k(\omega) \approx k(\omega_0) + \left. \frac{dk}{d\omega} \right|_{\omega_0} (\omega - \omega_0) = k(\omega_0) + k'(\omega - \omega_0).$$

Substituting into the expression:

$$e^{j[k(\omega)z - \omega t]} = e^{j[k(\omega_0)z + k'(\omega - \omega_0)z - \omega t + \omega_0 t - \omega_0 t]} = e^{j(k(\omega_0)z - \omega_0 t)} e^{j(\omega - \omega_0)(k'z - t)},$$

we get:

$$E(t, z) = e^{i(k(\omega_0)z - \omega_0 t)} \int_{-\infty}^{\infty} \tilde{E}(\omega) e^{j(\omega - \omega_0)(k'z - t)} d\omega.$$

This represents a carrier wave $e^{i(k(\omega_0)z - \omega_0 t)}$ modulated by an envelope that travels with velocity (by setting $k'z - t = \text{const}$ and taking the time-derivative of both sides):

$$v_g = \left. \frac{d\omega}{dk} \right|_{\omega_0} = \left(\left. \frac{dk}{d\omega} \right|_{\omega_0} \right)^{-1} = \frac{1}{k'}.$$

This group velocity v_g represents the speed at which the envelope—and hence energy and information—propagates through the medium.

For example, at high frequencies ($\omega > \omega_p$), metals become transparent as the refractive index n becomes real, and the phase velocity exceeds the speed of light in vacuum. However, when the group velocity is calculated (as shown in the homework), it is found to always remain below the speed of light in vacuum.

In conclusion, in dispersive media like metals described by the Drude model, phase velocity can exceed the speed of light, but it carries no information. The group velocity, derived from the dispersion relation $k(\omega)$, governs the transport of information and always remains subluminal.